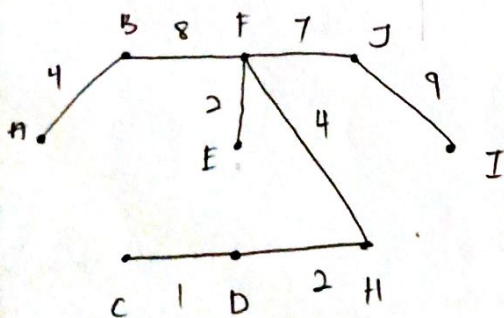


Question 4

a) Because vertex I does not have any edge to connect with and also the Kruskal's algorithm cannot have a circuit in it.

b)

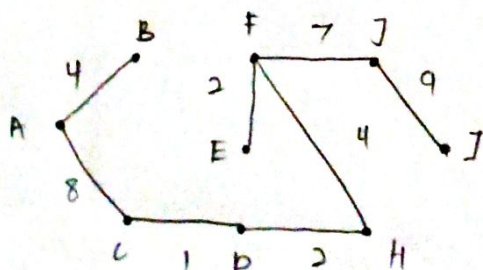
Edge	length	will adding edge make a new circuit	action taken	cumulative
$e_1 = (C, D)$	1	no	added	1
$e_2 = (D, H)$	2	no	added	3
$e_3 = (E, F)$	2	no	added	5
$e_4 = (F, H)$	4	no	added	9
$e_5 = (A, B)$	4	no	added	13
$e_6 = (E, D)$	6	Yes	not added	13
$e_7 = (E, C)$	7	Yes	not added	13
$e_8 = (F, J)$	7	no	added	20
$e_9 = (B, F)$	8	no	added	28
$e_{10} = (A, C)$	8	Yes	not added	28
$e_{11} = (J, I)$	9	no	added	37
$e_{12} = (J, H)$	10	Yes	not added	37



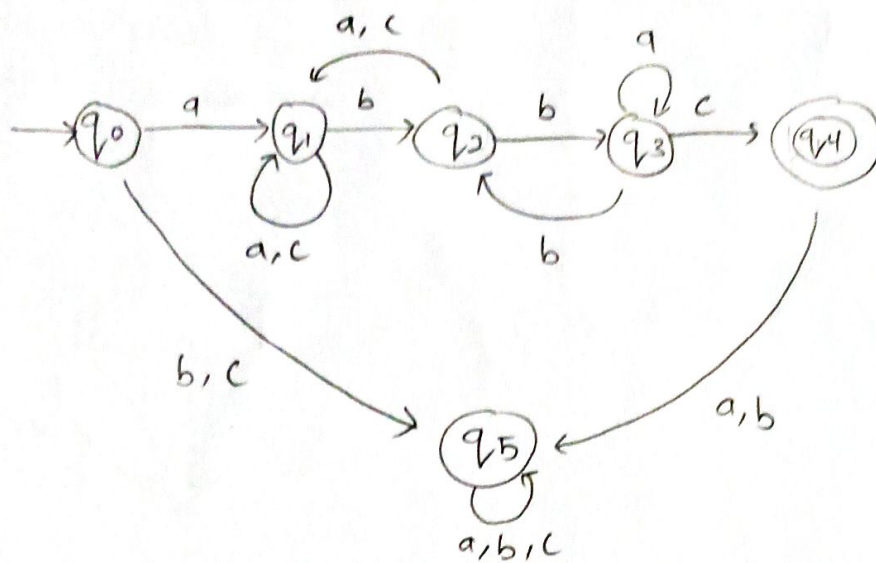
Length = 37 m

total cost = RM 3700

c) Yes, there are two edges that have the same length which are edge (B,F) and (A,C) and also both of the edges does not making a circuit if ~~we~~ we choose either one.



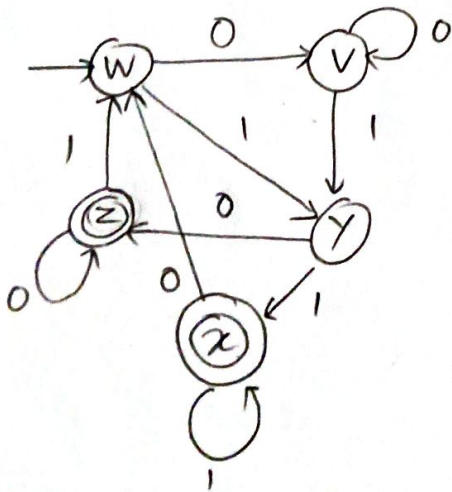
Question 5



Question 6

- a) state, $S = \{v, w, x, y, z\}$
 initial state = $\{w\}$

b)



$w \xrightarrow{0} v$	$v \xrightarrow{0} v$	$v \xrightarrow{1} y$	$y \xrightarrow{1} x$
$w \xrightarrow{0} v$	$v \xrightarrow{1} y$	$y \xrightarrow{1} x$	$x \xrightarrow{1} x$
$w \xrightarrow{0} v$	$v \xrightarrow{0} v$	$v \xrightarrow{1} y$	$y \xrightarrow{0} z$
$w \xrightarrow{0} v$	$v \xrightarrow{1} y$	$y \xrightarrow{0} z$	$z \xrightarrow{0} z$

Question 7

G = Ground floor

F1 = First floor

F2 = Second floor

